

Technical Implementation of IoT Concept for Bee Colony Monitoring

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Abstract— The modern autonomous beekeeping system developed in this research is the real example of Internet of Things technologies (IoT) in the beekeeping sector. It performs a bee colony maintenance control without interfering with its processes, while optimizing frequency of the apiary inspection. The system helps to analyze data correlation with video, meteo data, mass changes in time as well as interpretation of nest temperature, humidity and linking to local geographic and biological conditions. It allows a beekeeper to request and receive key data indicators and in accordance with the indicators to react on time and provide the best required maintenance of the bee colony. By implementing the autonomous beekeeping, the hives conditions can be tracked remotely, e.g. whether the inside temperature is critical, if the family is missing feed, therefore the critical deviation can be detected and prevented in time.

Keywords—autonomous beekeeping system, internet of things, data correlation, conditions, monitoring, control

I. INTRODUCTION

Precision Beekeeping (PB) is defined as an apiary management strategy based on monitoring of individual bee colonies to minimize the resource consumption and maximize the productivity of bees [11], [12],[13]. Open-source bee-hive monitoring system is developed by Glyn and Clive Hudson [14]. IoT system prototype for honey bee colony temperature monitoring is described in [15].

Nowadays a beekeeping is currently based largely on manual work that requires regular visits to bee apiaries and to monitor bee hives. However, physical inspection interferes bees' life and causes stress that negatively affects the productivity of all product lines. It is very important to find opportunities for remote monitoring of bee hive' profitability, flying activities, bee colony moods and bees' family health status. The physical indicators of this are: temperature and humidity controlling system, which is powered by solar energy. These indicators in their turn are to be controlled and monitored to provide the best

environment for bees according to the seasons, outside conditions, etc.

The proposed solution is the real Internet of Things technologies (IoT) example of implementation in the beekeeping sector. It will perform bee apiary control (individual bee colonies and at the apex level) without interfering with its processes, while optimizing frequency of the apiary inspection. The system will help to analyze data correlation with video, meteo data, mass changes in time as well as interpretation of nest temperature, humidity and linking to local geographic and biological conditions. It will allow a beekeeper to request and receive key data indicators. At the second stage of the project, the use of alternative energy sources (solar) for power supply will increase the application possibilities of the system at variety of geographical locations.

During a summer season beekeepers are forced to reallocate a bee apiary several time pursuing better conditions for harvesting of honey products. This fact sets requirement to the autonomous beekeeping system: wireless network which ensures data collection from sensors has to be easy and fast setup, independently on the electricity supply, reliable, safe and independent of weather conditions (rain, fog, froze, etc.)

The goal of the research is to test the quality and reliability of WSN network, which is used for bee hives monitoring system. This research explores the problems associated with vulnerability of Wireless Sensor Network, with the special focus on the weather conditions impact on the signal transmitting quality in WSN. The study of impact of weather conditions on the signal strength in outdoor wireless sensor networks, testing reliability of data transition, alerting a service performance degradation, the data transmitting joint concept in different weather conditions, based on the WSN testing results, was implemented in this research.

I. ARRANGEMENT OF AUTONOMOUS BEEKEEPING SYSTEM

A. System functionality and composition

At the project “Autonomous beekeeping”, the Riga Technical University in cooperation with two beekeeper farmers and Internet Association has created a unique autonomous beekeeping technology, based on IoT approach, used to monitor bee gardens remotely.

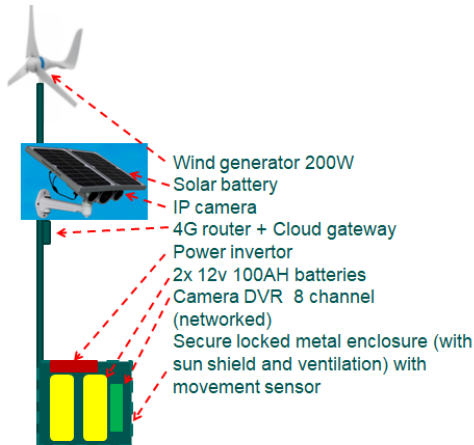


Fig. 1. Autonomous energy supply for smart monitoring.

We tested separate elements of the “Autonomous Beekeeping” system installed at five of bee hives in really field conditions – Riga Botanic Garden. During testing, the weight of bee hives, as well as the temperature sensors, which measure temperature ambient temperature and the temperature inside of the beehives, were monitored. Fig. 3 depicts the autonomous beekeeping systems organization scheme. Electrical power consumption of both common and individual elements of the equipment (Fig.2) was measured and monitored at the first stage.

B. Communication System

The validation of reliability of data transmission can be implemented by alerting a service performance degradation. The data transmitting in different weather conditions have been tested due to, it is an important part of the whole Autonomous beekeeping system.

In accordance with [4] the developed metering systems has to able to operate under diverse conditions: outdoor installation, indoor industrial installations; under electromagnetic pollution, temperature, humidity, rain impact, etc. Metering solution based on the modular principle for inner-system communication paradigm using selectable interface modules (IEEE 802.3, ISM radio interface, GSM/GPRS), having a focus on EMI electromagnetic interference impact at the proposed solution, which should be applied to the control of different critical infrastructure networks (water distribution and district heating networks, electricity network, etc.) using adapted modules.

The 4 G highly reliable and secure LTE router with I/O, GNSS and RS232/RS485 for professional applications was used for communication between gateway-concentrator and MQTT broker. The sustainability issues of autonomous beekeeping have been considered in [7, 8, 9] with the research focus on bees behavior in different environment condition. In our research we put emphasis on the development of a reliable monitoring system, which is resilient against changing weather conditions.

C. Video monitoring system

For more effective monitoring of the bee hives in addition to sensor data video cameras were used. The purpose of their use is to test the individual elements of the 'Autonomous Beekeeping' system under practical conditions, that is, one of bee apiaries. During the testing, it was planned to install video camera and video recording equipment on bee hives and perform video capabilities.

Two cameras were installed for video monitoring under real conditions:

- The whole view camera that allows you to monitor all hives
- Close view camera to observe the movement of the bees from / to one of the hives

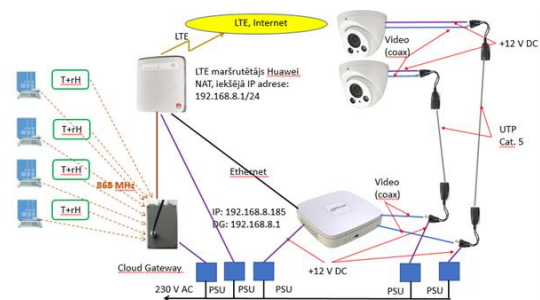


Fig. 2. Video monitoring block-scheme

The system is permanently in the internet connection.

The DVR and the Cloud Gateway sensor were connected to the Ethernet ports of a router providing also Wi-Fi for on-site work - to receive video streams and to regulate / administer / configure the DVR. For an easier administration the DVR was set to a static address. The Mobile Internet LTE service provided Internet access via Network Address Translation (NAT) and Port Address Translation (PAT) by a mobile operator's firewall, and internal (non-routable) IP addresses are assigned to the LTE router's external interface. The video monitoring provides two views – general view and individual at particular beehive.

D. Data collection and interpretation

The beehive monitoring system deployed a Node-RED [14] as a gateway concentrator for wiring together hardware devices (temperature, weight, humidity sensors), APIs and online services. Node-RED main functionality is to decode and to route MQTT smart metering data to further service orchestration or use in external services as monitoring system or external clients.

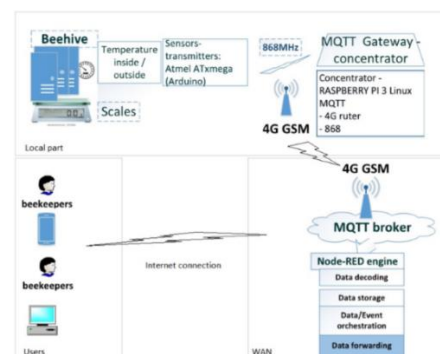


Fig. 3. Test bed layout: beehive monitoring system.

There are 3 surface sensors-transmitters for measuring of temperature inside hive in 3 different locations, in the center and sides (Fig. 4) and outside the stack (Fig. 5) to get the outdoor air temperature are used for one beehive. The weight of each hive is measured by specially designed weight platform (Fig.5) and are sent by a sensor-transmitter to the concentrator without processing.



Fig. 4. Measuring of temperature inside hive.

The measurements from sensor-transmitters are transmitted to concentrator from sensors using ISM range signal. The signals from the sensors are processed and analyzed for further use.



Fig. 5. Outdoor temperature measurement.

The temperature sensor-transmitters transfer data to the server-broker every 15 minutes, the weight sensor transmits data to the server every 5 minutes. All data collected on the MQTT broker are displayed in graphical form. The users interpret data for understanding of ongoing processes in the hive.

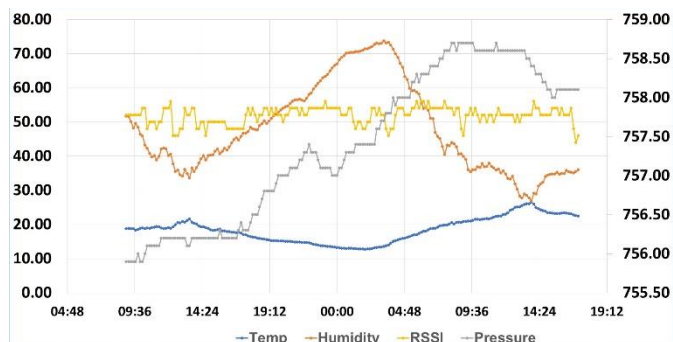


Fig. 6. Environment factor and RSSI Correlation.

The weather impact on the tested system was assessed, by comparison RSSI data and the weather conditions data from the

Weather monitoring system of Botanical garden in Riga. Environment factor and RSSI correlation during the summer period was analyzed. The weather impact, measured during summer period, was used as a reference for evaluation of Quality of Delivery (QoD). A snapshot of changing weather conditions is presented at Fig. 6.

The actual topology of WSN was defined based on the energy consumption level of nodes, distances between bee hives location points, and weather conditions at defined geographical location in order to assure reacquired level of Quality of Delivery (QoD). It is noticed that the highest impact at the RSSI data is provided by humidity changes, but less impact – temperature and pressure variations. However, the system showed fair sustainability, due to RSSI data deviation happened within the normal range.

In order to process and interpret results a prototype of MQTT broker and the data collection and transmission system was tested as a part of autonomous beehives system. For further data processing, monitoring and interpretation an Emoncms code base is used. Emoncms represents measurement data as a dashboard and as the charts. The print screens show the temperature inside and outside of beehives, and deviations of beehives weight (see Fig.7). Three of the graphs refer to temperature inside and outside the hive. Given that earlier it is stated that there are 3 sensors installed at different heights inside the hive, please explain which graph corresponds to each sensor. Two charts at the left side show the temperature inside a hive. The right upper chart shows the temperature outside the hive, but the right bottom chart shows the weight of the beehive. All measurements at Fig. 7 were made in August 2018.

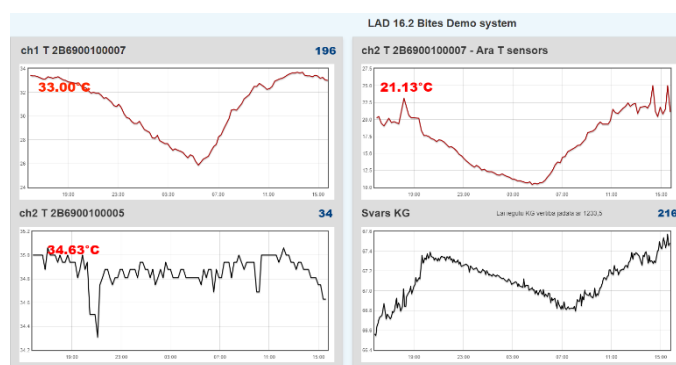


Fig. 7. Web based interface of on-line published monitoring results. Measurements of the temperature inside and outside of beehives, and weight deviations.

It was noticed that the bees maintain a constant ecosystem within $\pm 0.5^{\circ}\text{C}$ at the center of the beehive. The absolute mean value is 34.8°C that really differs from the classical assumption.

Beekeepers, the partners of the reach are interested to compare behavior of bee colonies and particularly, at the transitions of the seasons, when bees start to awake, to leave hives after winter time. The new version of AB provides opportunity of comparison of bee colonies thanks to multi-chart view (see Fig.8). Beekeepers, who are the partners of this research, are interested to compare behavior of bee colonies and particularly, at the transitions of the seasons, when bees awake and start to leave their hives after winter time. The new version of AB system, which was developed by the end the summer season of 2018, provides opportunity to compare behavior of bee colonies

thanks to multi-chart view (see the chart that depicts temperature deviations at Fig.8).

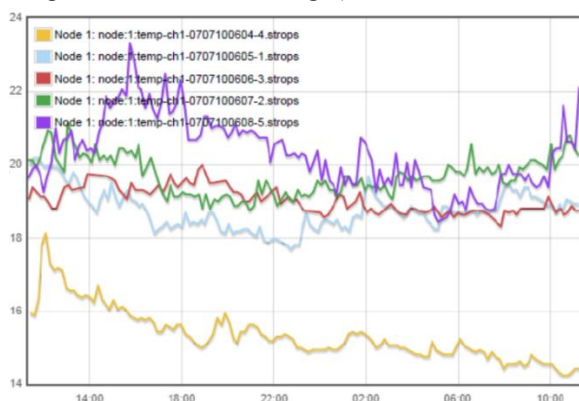


Fig. 8. Multi-chart view of temperature measurement inside of 5 beehives

During the summer season additionally sensors were installed at the other 4 beehives. The measurements at Fig.8 were made at the end of summer season, when the ambience temperature and accordingly the temperature inside of hives were decreasing.

II. CONCLUSIONS

Development of autonomous beekeeping system is still in progress, since the project team implemented the first stage of the project. Recent achievement of IoT provide the opportunities for Implementation of its principles in autonomous beekeeping for creation reliable system, which is resilient to the weather conditions.

The offered autonomous beekeeping system applies sensors that provide useful data on hive status (internal and ambient temperature, humidity, weight of the hives). Such sensors provide important data to the users, so they can evaluate the hive and take further action.

3G technology is used to transmit the sensor data from the hive to the user's phone or computer, because it provides stable, secure and relatively fast data transfer. Furthermore, availability of GSM connection in all parts of Latvia promotes its application at autonomous beekeeping system.

Meteo-data were used to evaluate correlation between measured weather parameters and deviation of the parameters, which characterize QoS of WSN. The data quality was assessed based on experimental data. The quality and reliability of WSN network were tested using bee hives system's monitoring tools.

It has been concluded that the described system for the monitoring of beehives can be improved with additional sensors and various actuators. At the second stage of the project video monitoring system will be introduced as a part of the system. Additional research is planned in order to ensure autonomous energy supply, so it is necessary to evaluate consumption of energy and opportunity to apply alternative energy sources, such wind and photovoltaic elements. The highest impact at the RSSI data is provided by humidity changes, but less impact – temperature and pressure variations.

The developed solution could be considered as TRL 5 level, however, for the future commercial use, the data reliability is recognized as one of the critical points. The measurement accuracy and data quality from the testing network are in line with the initial forecasts. Interestingly, the frequent reading of weight data opens the way for analyzing the ongoing steps at a more detailed level than originally anticipated. The use of intelligent bee hives increases bee productivity beekeepers work easier. The further plans for the work are to estimate how the developed intelligent beehives and installed embedded system increase productiveness of the beekeepers work not disturbing the harmony of the bee colony system.

ACKNOWLEDGMENT

This article is based upon the project “Autonomous Beekeeping” that is funded by the European Agricultural Fund for Rural Development Program, 2014-2020, the project is initiated by Riga Technical University, Latvian Internet Association and two beekeeper farmers in the beginning 2018, Cooperation: Support new products, methods, processes and technologies.

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